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1) Reduction of chloroform

a) Reduction of chloroform in the presence of Zn + HCl gives methylene chloride.

$$\underset{\text{chloro form}}{CHCl_3} \xrightarrow[2(H)]{Zn+HCl} \underset{\text{methylene chloride}}{CH_2Cl_2 + HCl}$$

b) Reduction of chloroform using H2/Ni

$$\underset{\text{chloro form}}{CHCl_3} \xrightarrow[Ni]{H_2} \underset{\text{methylene chloride}}{CH_2Cl_2 + HCl}$$

2) Chlorination of methane

Chlorination of methane gives methylene chloride

$$\underset{\text{Methane}}{CH_4} \xrightarrow[-HCl]{Cl_2/h\upsilon} \underset{\text{Methyl chloride}}{CH_3Cl}$$

$$\xrightarrow[-HCl]{Cl_2/h\upsilon} \underset{\text{Methylene chloride}}{CH_2Cl_2}$$

Uses of methylene chloride

Methylene chloride is used as

1) aerosol spray propellant
2) solvent in paint remover
3) process solvent in the manufacture of drugs
4) a metal cleaning solvent

14.6.2 Trihaloalkane

Trihaloalkanes are compounds obtained by replacing three hydrogen atoms of a hydrocarbon by three halogen atoms.

Example

$CHCl_3$	CHI_3
Chloroform	Iodoform

1) Chloroform

Chloroform is an important trihaloalkane. Dumas named $CHCl_3$ as chloroform as it gives formic acid on hydrolysis.

Preparation:

Chloroform is prepared in the laboratory by the reaction between ethyl alcohol with bleaching powderfollowed by the distillation of the product chloroform. Bleaching powder act as a source of chlorine and calcium hydroxide. This reaction is called haloform reaction. The reaction proceeds in three steps as shown below.

Step – 1: Oxidation

$$CH_3CH_2OH + Cl_2 \rightarrow CH_3CHO + 2HCl$$

Ethyl alcohol Acetaldehyde

Step – 2: Chlorination

$$CH_3CHO + 3Cl_2 \rightarrow CCl_3CHO + 3HCl$$

Acetaldehyde Trichloro acetaldehyde

Step – 3: Hydrolysis

$$2CCl_3CHO + Ca(OH)_2 \rightarrow 2CHCl_3 + (HCOO)_2Ca$$

Chloral chloroform